

## Entropy Generation Analysis on MHD Ag+Cu/Blood Tangent Hyperbolic Hybrid Nanofluid Flow Over a Porous Plate

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**ABSTRACT:** This study is motivated by the vital role of dissipating thermal energy in the physiological system where energy depletion can lead to severe health complications. These complications encompass a range of issues, including sudden death, anemia, hypothermia, blood pressure fluctuations and the necessity for cardiac surgery. Biomedical engineers and clinicians have recognized the significance of analyzing entropy generation to quantify energy loss in biological systems. Furthermore, this study acknowledges the importance of understanding the thermodynamic state of entropy generation, particularly in evaluating cancer cells during chemotherapy treatment and enhancing heat transfer in tissues. The primary objective of this study is to evaluate the heat transfer characteristics of a magnetohydrodynamic (MHD) tangent hyperbolic hybrid nanofluid near a heat source and thermal radiation as it flows over a porous plate. The research methodology utilizes the MATLAB program bvp4c for solving the momentum and temperature equations. These equations are subsequently transformed into ordinary differential equations using the appropriate self-similarity variables. An elevation in the heat source parameter leads to heightened internal energy of liquid particles, resulting in an increase in temperature. Additionally, the magnetic field parameter is directly proportional to the entropy generation; as it increases, so does the entropy generation. Moreover, nanoparticles, owing to their high surface area-to-volume ratio, have the capacity to hinder heat transport within the fluid. The specific application of this study lies in the field of biomedical engineering and clinical practices. The findings can contribute to developing advanced heat-transfer techniques for medical applications, such as improving chemotherapy treatments for cancer cells and enhancing tissue heat-transfer efficiency. Moreover, using silver and copper nanoparticles as heat-transfer agents could hold promise in treating blood-related health conditions and facilitating the healing of injured tissue.

**KEYWORDS:** Entropy generation; MHD; thermal radiation; porous plate; hybrid nanofluid.

### NOMENCLATURE

|            |                                      |
|------------|--------------------------------------|
| We         | Weissenberg number                   |
| Br         | Brinkman number                      |
| Gr         | Grashof number                       |
| $k'$       | Permeability of the porous medium    |
| $n_a$      | Power-law index                      |
| $\gamma$   | Half angle (rad)                     |
| $T_\infty$ | Temperature of the ambient fluid (K) |
| $T_w$      | Wall temperature (K)                 |

|        |                                                                       |
|--------|-----------------------------------------------------------------------|
| $M$    | Magnetic field parameter                                              |
| $Re_x$ | Local Reynolds number                                                 |
| $Q_0$  | Heat generation/absorption coefficient                                |
| $c_p$  | Specific heat at constant pressure ( $\text{Jkg}^{-1}\text{K}^{-1}$ ) |
| $T$    | Temperature of the fluid (K)                                          |

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|           |                                                 |
|-----------|-------------------------------------------------|
| $k^*$     | Mean absorption coefficient ( $\text{m}^{-1}$ ) |
| $k$       | Thermal conductivity                            |
| $N_r$     | Nusselt number                                  |
| $C_f$     | Skin friction coefficient                       |
| $u$ & $v$ | Velocity components ( $\text{m/s}$ )            |
| $x$ & $y$ | Cartesian coordinates ( $\text{m}$ )            |
| $Q$       | Heat source parameter                           |
| $R_d$     | Radiation parameter                             |
| $Ec$      | Eckert number                                   |
| $K$       | Porosity parameter                              |
| $Pr$      | Prandtl number                                  |

### Greek symbols

|            |                                                               |
|------------|---------------------------------------------------------------|
| $\rho$     | Density ( $\text{kg m}^{-3}$ )                                |
| $\beta_T$  | Coefficient of thermal expansion ( $\text{m}^3/\text{kmol}$ ) |
| $\phi$     | Volume fraction of nanoparticle                               |
| $\alpha_1$ | Temperature difference parameter                              |
| $\theta$   | Dimensionless temperature                                     |
| $\mu$      | Dynamic viscosity ( $\text{kg m}^{-1} \text{s}^{-1}$ )        |
| $\eta$     | Similarity variable                                           |
| $\nu$      | Kinematic viscosity ( $\text{m}^2\text{s}^{-1}$ )             |

### Subscripts

|       |                  |
|-------|------------------|
| $hnf$ | Hybrid nanofluid |
| $bf$  | Base fluid       |
| $f$   | Fluid            |

### Superscript

|   |                                          |
|---|------------------------------------------|
| ' | Differentiation with respect to $\eta$ . |
|---|------------------------------------------|

## 1. INTRODUCTION

The phenomenon of fluid flow along a stationary surface holds great importance in the field of fluid mechanics, initially introduced by Blasius. Sakiadis expanded the scope of this issue to encompass the flow of fluid over a dynamic surface. Owing to their many uses in manufacturing and engineering, which include plastic extrusion, uninterrupted casting, glass fiber and crystal formation, such flow problems have piqued the interest of researchers. A lot of studies used various types of physical impacts and procedures to describe the Blasius and Sakiadis idea.<sup>1-4</sup> In the past, plate heat-transfer systems were used extensively in the food and beverage industries, as they can be readily taken apart for periodical inspection and cleaning. Due to their features and advantages, plate heat exchangers in the last few years, have started continuous expansion of chemical fields in nature, process and allied industries. Plate heat exchangers have increasingly proven to be useful in a wide range of industrial applications, i.e. for

single phase heat-transfer processes as heater or cooler as well as for condensation and evaporation. Wherever possible, a plate heat exchanger will be usually an economic solution to fulfil the required heat-transfer process, subject to safety and reliability considerations. Many industries are now turning to fixed or moving bed heat exchangers (MBHEs) that use vertical plate technology to indirectly heat and cool material via conduction across the surface of vertical steel plates. This technology, which can also be used to dry solids, has shifted the framework of what is possible in both traditional and nontraditional bulk solids industries. MBHEs that use vertical plate technology allow the indirect heat transfer between solids and fluids. Plate and frame exchangers are the most efficient heat-transfer mechanisms found in the industry globally. Water heaters, tower cooling isolation, direct cooling, heat from waste retrieval, devices for thermal (ice) storage and heat pump isolation are examples of industries where this high efficiency is visible. Ahmed *et al.*<sup>5</sup> examined the magnetohydrodynamic (MHD) nanofluid flow across an inclined plate in an inclined, semi-infinite, nonreflecting and perfectly transparent porous medium. Their findings indicated that increasing the inclination angle of the plate resulted in a corresponding increase in the thermal buoyancy effect. Consequently, this led to enhanced velocity profiles and higher maximum temperature in the system. Zahrani *et al.*<sup>6</sup> conducted a research study with the objective of improving the comprehension of the thermal performance of the conventional flat plate heat exchanger (FPHE) and devising an enhanced iteration known as the modified FPHE. The researchers proposed a new flow arrangement with the aim of improving the thermal efficiency of the FPHE. Considering that the Fanning friction factor ( $f$ ) of the basic FPHE is small, the proposed modification to the flat plate's surface shows great promise, as it results in an acceptable increase in the Fanning friction factor while still maintaining performance within an acceptable range. Mahabaleshwar *et al.*<sup>7</sup> emphasized that the use of mathematical modeling, particularly involving nanofluid nanoparticles and the porous medium approach, has become a contemporary method for gaining insights into cancer cell treatment processes.

MHD has piqued the interest of many researchers in recent decades due to its wide range of applications in a variety of industries. Polymer processes, plastic sheet extrusion, X-ray radiation, vehicle cooling, fiber filters, electrolytes, magnetic cells, loudspeaker manufacturing and so on are examples of such uses. MHD is a field of study within the realm of physical science that pertains

to the behavior of electrically conductive fluids in the presence of a magnetic field. Magneto-fluid dynamics is another name for this phenomenon. Electrolytes, plasmas, saltwater and liquid metals are examples of magneto-fluids. In 1970, Hannes Alfvén, who pioneered the research of MHD, was awarded the Physics Nobel Prize. MHD is employed in many technical and industrial applications, including generators, electronic filters, power pumps, reactor cooling, power generation, the bloodstream, MHD detectors, magnetic targeting of drugs, crystal growth and many more. Numerous scholars and specialists express interest in analyzing the impact of magnetic fields on behavior, particularly with regard to heat transfer considerations. Typically, researchers utilize liquid metals, water with a small amount of acid and other industrial fluids in the presence of a magnetic force to analyze the flow and thermal conduction of a viscous and incompressible liquid as it passes over various geometries. Rashed *et al.*<sup>8</sup> investigated the primarily concerned magnetic field by analyzing the free convective flow and heat transfer of nanofluid near an inclined plate submerged in a porous media. The findings indicated that an escalation of magnetic field parameters resulted in a significant decrease in the maximum tangential velocity, while the maximum temperature within the flow field increased. Reddy *et al.*<sup>9</sup> conducted an analysis of producing entropy in a microchannel with electrokinetic force and electroconductive heating, focusing on the flow of gold-blood pseudoplastic nanofluid. The findings of their study indicate that the magnetic field has a decreasing effect on the velocity of pseudoplastic blood flow in the central region of the channel. Reddy<sup>10</sup> studied the effects of heat radiation, viscosity dissipation and the Lorentz force on the movement of blood-carrying titanium alloy-gold hybrid nanoparticles in a revolving tube. The results show that when the magnetic field intensity increases, the velocity and temperature of the blood-based nanofluid drop. Jakeer *et al.*<sup>11</sup> investigated the effects of the produced magnetic field influence on Eyring-Powell/Maxwell hybrid nanofluid flow in a linear expanding sheet with non-Fourier heat flux and entropy generation. A micropolar fluid's MHD heat transfer flow through a vertical porous moving plate with radiation effects was examined by Bejawada *et al.*<sup>12</sup> Numerical solutions obtained using the Galerkin finite element technique revealed that higher magnetic field parameter values lead to an increase in skin friction. Chu *et al.*<sup>13</sup> investigated the double diffusion impact of a stretchy nanomaterial with Arrhenius activation energy on bio-convective flow of magneto-polymer tangent hyperbolic (non-Newtonian) liquid.

During the time of the machine, a new technique was invented a few years ago for the purpose of implementing very small particles into the base fluid. This approach aims to enhance the machinery and improve the chilling system of these devices. Nanofluids are utilized to improve the base fluid's efficiency of heat transfer and thermal conductivity. Triethylene glycol, water, ethylene, oil-based chemicals and oils, polymeric solutions and biofluids are examples of basic liquids that are often used. Other examples include biofluids. Nanoparticles made from copper, silver, aluminum, titanium dioxide, zirconia, aluminum oxide, copper oxide, aluminum nitride, silicon nitride, aluminum oxynitride, aluminum oxynitride, graphite, diamond and carbon nanotubes are all widely employed. Nanofluids are special because of their uniformity high ability to conduct heat at low-nanoparticle aggregation, long-term stability and minimal tendency to clog up flow pathways. Because of this, these fluids are used in an extensive range of electrical products, including fluidic digital display devices micromechanical systems, as well as chip cooling. Nanofluids find extensive applications across diverse domains, encompassing but not restricted to the following: Thermal management in buildings, heat exchange processes, transportation sector, sensing technologies, microfluidic systems, lubrication mechanisms, pharmaceutical production, nanocryosurgical procedures, cooling of electronic devices, cancer therapy, cryogenic preservation, imaging techniques and preservation of biological specimens. Despite the benefits of nanofluids, there are still some drawbacks, such as the fact that just one kind of nanoparticle cannot improve all of the base fluid's necessary qualities. Hybrid nanofluids have been developed to address this issue. By adding more than one nanoparticle to the basic fluid, hybrid nanofluids are created. The physical and chemical characteristics of many materials come together simultaneously in hybrid nanofluids to create a homogeneous mixture. By increasing thermal conductivity and the heat transfer rate, these nanofluids have decreased costs in industrial domains. A review of this subject can be found in Refs. 14–18. Silver nanoparticles stand out among metal nanoparticles because of their exceptional optical, electrical and thermal capabilities. They have excellent thermal conductivity (429 W/mK at 300 K) and are hence expected to have outstanding heat transfer qualities, making them ideal for thermal applications. Because of its outstanding properties such as quantum confinement, large surface area and minuscule diameter, silver nanoparticles play an incredibly important role in a wide range of sectors. Nanoparticles

have been employed as antibacterial agents in various settings, including factories, households and health-care-related products. Additionally, they have found applications in household goods, medical equipment coatings, optical sensors and beauty products. In the pharmaceutical industry, nanoparticles have been utilized in drug delivery systems, diagnostics and orthopaedics. Notably, their unique properties have been leveraged to enhance the efficacy of anticancer drugs, thereby augmenting their tumor-killing effects. Khan and Arasu<sup>19</sup> conducted an investigation on the synthesis of TiO<sub>2</sub> with Ag nanoparticles in composite form, followed by their suspension in water, to produce TiO<sub>2</sub>-Ag/water hybrid nanofluids. Reddy<sup>20</sup> examination was undertaken on a time-dependent pulsatile MHD hybrid nanofluid in a channel. The investigation focused on entropy optimization and the influence of electrokinetic force. The results of the study suggest that an augmentation in the velocity of the blood-based Cu-Au hybrid nanoparticle volume fraction results in an elevation in entropy generation. Nabwey *et al.*<sup>21</sup> used a hybrid nanofluid (Ag+Cu-water) to describe how radiation and heat generation/absorption influence an unstable flow in an angled crimped porous cavity. The findings indicated higher nanoparticle volume fractions led to an increased average Nusselt number.

This study explores the novel and distinctive aspects of MHD tangent hyperbolic hybrid nanofluids concerning their heat transfer characteristics in the occurrence of a porous plate with a heat source and thermal radiation, while also considering entropy generation. Previous literature in this area is limited, indicating a research gap that this study aims to address.

- How does the Grashof number affect the velocity profile of MHD tangent hyperbolic hybrid nanofluids near a porous plate, and what is the relationship between the Grashof number and the velocity?
- What is the impact of the Weissenberg number on the velocity profile of MHD tangent hyperbolic hybrid nanofluids, and how does the velocity change with increasing values of Weissenberg number?
- How does the rate of temperature rise in MHD tangent hyperbolic hybrid nanofluids vary with higher values of heat source and radiation parameters?
- How does the volume percentage of nanoparticles influence the rate of entropy generation in MHD tangent hyperbolic hybrid nanofluids, and what is the relationship between nanoparticle volume percentage and the rate of entropy generation?

- What is the impact of varying the volume fraction of nanoparticles and radiation parameters on the rate of heat transfer in MHD tangent hyperbolic hybrid nanofluids? In what manner do these parameters contribute to the augmentation of heat transfer rate within the system?

To tackle these research questions, this study employs the MATLAB program bvp4c to solve momentum and thermal equations, contributing to the understanding of the heat transfer and entropy generation processes.

## 2. THE MATHEMATICAL EXPLANATION OF THE PROBLEM

The expression for the stress tensor of a liquid under the influence of tangent hyperbolic forces is given by

$$\tilde{\tau} = \tilde{\omega} \left[ \zeta_{\infty} + (\zeta_{\infty} + \zeta_0) \tanh(\Gamma \tilde{\omega})^{n_a} \right].$$

In this context,  $\zeta_{\infty}$  represents the viscosity at infinite shear rate,  $\Gamma$  denotes the time constant,  $\zeta_0$  signifies the viscosity at zero rate,  $\tilde{\tau}$  represents the additional stress tensor and  $\tilde{\omega}$  is defined as follows:

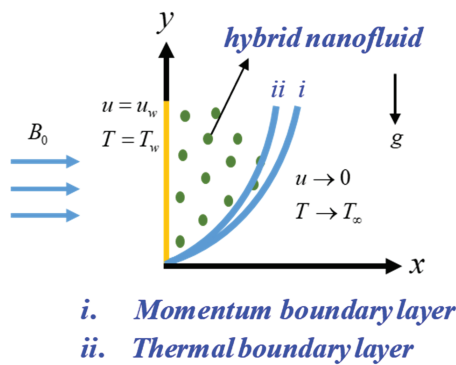
$$\tilde{\omega}^2 = \frac{1}{2} \left( \sum_i \sum_j \tilde{\omega}_{ji} \tilde{\omega}_{ij} \right) = \frac{\Pi}{2}.$$

Here  $\Pi = \frac{1}{2} \text{tr}((\nabla \vec{V})^T + (\nabla \vec{V}))^2$ . Let us assume  $\zeta_{\infty} = 0$ . Our focus is directed towards the investigation of the shear thinning characteristics exhibited by  $\tilde{\omega}\Gamma < 1$  and  $\tilde{\omega}$  is reduced as follows:

$$\tilde{\tau} = \left[ \left( (\tilde{\omega}\zeta)^{n_a} \right) \right] \zeta_0 \tilde{\omega} = \left[ 1 + n_a (\tilde{\omega}\Gamma - 1) \right] \zeta_0 \tilde{\omega}.$$

This study investigates the behavior of an incompressible tangent hyperbolic hybrid nanofluid as it flows over a given geometry, taking into account the effects of thermal radiation and heat generation. The rheological properties of blood render it to display attributes akin to those of a tangent hyperbolic fluid, wherein copper and silver nanoparticles are suspended. The consideration of viscous dissipation and joule heating is incorporated into the analysis. The  $x$ -axis is situated parallel to the surface of the body, while the  $y$ -axis is understood to be perpendicular to the geometry's surface. Furthermore, the presence of a magnetic field  $B_0$  is taken into account, which is oriented perpendicular to the geometric configuration. Figure 1





**Fig. 1.** (Color Online) Modeling of the framework and coordinate system.

presents a representation of the modeling framework.  $T_w$  is the wall temperature of the geometry,  $T_\infty$  is the temperature far away from it. In addition, the temperature at the surfaces and exterior the boundary is  $T_w$  at ( $y = 0$ ) and  $T_\infty$  ( $T_\infty < T_w$ ) as ( $y \rightarrow \infty$ ) and  $M^*$  is the geometric factor. On the basis of these assumptions presented, the equations that are used to determine things like continuity, momentum and temperature are as follows<sup>22–24</sup>:

$$\frac{\partial(r^{M^*}u)}{\partial x} + \frac{\partial(r^{M^*}v)}{\partial y} = 0, \quad (1)$$

$$\begin{aligned} u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} = & \frac{\mu_{hnf}}{\rho_{hnf}} \frac{\partial^2 u}{\partial y^2} \left( (1-n_a) + \sqrt{2}n_a \Gamma \frac{\partial u}{\partial y} \right) \\ & + g(T - T_\infty) \beta_{T_{hnf}} \cos \gamma - \frac{\mu_{hnf}}{\rho_{hnf}} \frac{u}{K^*} \\ & - \frac{\sigma_{hnf}}{\rho_{hnf}} B_0^2 u, \end{aligned} \quad (2)$$

$$\begin{aligned} u \frac{\partial T}{\partial x} + v \frac{\partial T}{\partial y} = & \frac{k_{hnf}}{(\rho C_p)_{hnf}} \frac{\partial^2 T}{\partial y^2} + \frac{\mu_{hnf}}{(\rho C_p)_{hnf}} \left( \frac{\partial u}{\partial y} \right)^2 \\ & \left( (1-n_a) + \frac{n_a \Gamma}{\sqrt{2}} \frac{\partial u}{\partial y} \right) - \frac{1}{(\rho C_p)_{hnf}} \frac{\partial q_r}{\partial y} \\ & + \frac{\sigma_{hnf} B_0^2}{(\rho C_p)_{hnf}} u^2 + \frac{Q_0}{(\rho C_p)_{hnf}} (T - T_\infty). \end{aligned} \quad (3)$$

The pertinent boundary conditions are outlined as follows:<sup>25</sup>

$$\begin{aligned} u = u_w = \frac{v_f x}{l^2}, \quad v = 0, \quad T = T_w \quad \text{at } y = 0, \\ u \rightarrow 0, \quad T \rightarrow T_\infty \quad \text{as } y \rightarrow \infty. \end{aligned} \quad (4)$$

The Rosseland technique is used to calculate the radiative thermal flux.<sup>25</sup>

$$q_r = -\frac{4\sigma^*}{3k^*} \frac{\partial T^4}{\partial y}. \quad (5)$$

Considering the observed temperature fluctuations within the stream, it is evident that  $T^4$  can be mathematically represented as a linear combination of the variances. By extending the term  $T^4$  up to time  $T_\infty$ , the subsequent outcomes are as follows<sup>25</sup>:

$$T^4 = T_\infty^4 + 4T_\infty^3(T - T_\infty) + 6T_\infty^2(T - T_\infty)^2 + \dots \quad (6)$$

It is possible to derive the following by removing higher order parts from Eq. (6) that are greater than the first degree in  $T - T_\infty$ :

$$-3T_\infty^4 + 4TT_\infty^3 \cong T^4. \quad (7)$$

By employing Eqs. (5) and (7), the following is obtained:

$$\frac{\partial q_r}{\partial y} = -\frac{\partial^2 T}{\partial y^2} \frac{16A^*}{3k^* T_\infty^3}. \quad (8)$$

The following list of similarity variables is presented<sup>25</sup>:

$$\begin{aligned} u = \frac{v_f x}{l^2} f', \quad v = \frac{-v_f (M^* + 1)}{l} f, \\ \theta = \frac{T - T_\infty}{T_w - T_\infty}, \quad \eta = \frac{y}{l}. \end{aligned} \quad (9)$$

Using nondimensional parameters, simplified equations are described as follows:

$$\begin{aligned} A_1 f''' \left( (1-n_a) + n_a Wef'' \right) \\ + A_2 \left( (M^* + 1) ff'' - f'^2 \right) \\ + A_3 Gr \theta \cos \gamma - KA_1 f' - A_4 Mf' = 0, \end{aligned} \quad (10)$$

$$\begin{aligned} \frac{1}{Pr} \left( \frac{4}{3} Rd + A_6 \right) \theta'' + A_5 (M^* + 1) f \theta' + A_1 Ec f'^2 \\ \left( (1-n_a) + \frac{Wen_a}{2} f'' \right) + Q\theta + A_4 MEcf'^2 = 0. \end{aligned} \quad (11)$$

Boundary conditions are accompanied by

$$\begin{aligned} f(0) = 0, \quad f'(0) = \theta(0) = 1 \quad \text{at } \eta = 0, \\ f'(\infty) \rightarrow 0, \quad \theta(\infty) \rightarrow 0 \quad \text{as } \eta \rightarrow \infty. \end{aligned} \quad (12)$$

Here

$$A_1 = \frac{\mu_{hnf}}{\mu_f}, A_2 = \frac{\rho_{hnf}}{\rho_f}, A_3 = \frac{(\rho\beta_T)_{hnf}}{(\rho\beta_T)_f},$$

$$A_4 = \frac{\sigma_{hnf}}{\sigma_f}, A_5 = \frac{(\rho c_p)_{hnf}}{(\rho c_p)_f}, A_6 = \frac{k_{hnf}}{k_f}.$$

The following are the thermophysical properties of hybrid nanofluids<sup>26</sup>:

$$\frac{\mu_{hnf}}{\mu_f} = \frac{1}{(1 - \phi_{Cu})^{2.5} (1 - \phi_{Ag})^{2.5}},$$

$$\frac{(\rho\beta_T)_{hnf}}{(\rho\beta_T)_f} = (1 - \phi_{Ag}) \left[ (1 - \phi_{Cu}) + \phi_{Cu} \frac{(\rho\beta_T)_{Cu}}{(\rho\beta_T)_f} \right] + \phi_{Ag} \frac{(\rho\beta_T)_{Ag}}{(\rho\beta_T)_f},$$

$$\frac{\rho_{hnf}}{\rho_f} = (1 - \phi_{Ag}) \left[ (1 - \phi_{Cu}) + \phi_{Cu} \frac{\rho_{Cu}}{\rho_f} \right] + \phi_{Ag} \frac{\rho_{Ag}}{\rho_f},$$

$$\frac{(\rho c_p)_{hnf}}{(\rho c_p)_f} = (1 - \phi_{Ag}) \left[ (1 - \phi_{Cu}) + \phi_{Cu} \frac{(\rho c_p)_{Cu}}{(\rho c_p)_f} \right] + \phi_{Ag} \frac{(\rho c_p)_{Ag}}{(\rho c_p)_f},$$

$$\frac{\sigma_{hnf}}{\sigma_{bf}} = \frac{\sigma_{Ag} + 2\sigma_{bf} - 2\phi_{Ag}(\sigma_{bf} - \sigma_{Ag})}{\sigma_{Ag} + 2\sigma_{bf} + \phi_{Ag}(\sigma_{bf} - \sigma_{Ag})},$$

where  $\frac{\sigma_{bf}}{\sigma_f} = \frac{\sigma_{Cu} + 2\sigma_f - 2\phi_{Cu}(\sigma_f - \sigma_{Cu})}{\sigma_{Cu} + 2\sigma_f + \phi_{Cu}(\sigma_f - \sigma_{Cu})},$

$$\frac{k_{hnf}}{k_{bf}} = \frac{(1 + 2\phi_{Ag})k_{Ag} + 2(1 - \phi_{Ag})k_{bf}}{(1 - \phi_{Ag})k_{Ag} + (2 + \phi_{Ag})k_{bf}},$$

where  $\frac{k_{bf}}{k_f} = \frac{(1 + 2\phi_{Cu})k_{Cu} + 2(1 - \phi_{Cu})k_f}{(1 - \phi_{Cu})k_{Cu} + (2 + \phi_{Cu})k_f}.$

From Eqs. (10) and (11),

$We = \sqrt{2} \frac{u_w \Gamma}{l}$  is the Weissenberg number,  $K = \frac{l^2}{K^*}$  is the porosity parameter,  $M = \frac{\sigma_f B_0^2 l^2}{\mu_f}$  is the magnetic parameter,  $Pr = \frac{\mu_f c_{p_f}}{k_f}$  is the Prandtl number,  $Rd = \frac{4\tilde{A}^* T_\infty^3}{k^* k_f}$  is the radiation parameter,  $Ec = \frac{u_w^2}{c_{p_f}(T_w - T_\infty)}$  is the Eckert number,  $Q = \frac{Q_0 l^2}{\mu_f c_{p_f}}$  is the heat generation,

$\alpha_1 = \frac{\Delta T}{T_\infty}$  is the temperature difference parameter,  $Br = \frac{\mu_f u_w^2}{\Delta T k_f}$  is the Brinkman number and  $Gr = \frac{g \beta_f (T_w - T_\infty) l^2}{u_w \nu_f}$  is the Grashof number. We can obtain a local similar solution by utilizing the similarity transformation (Eq. (9)). As a result, the parameters Weissenberg number, Eckert number, Brinkman number and Grashof number are considered to be local dimensionless parameters. At a particular position on the sheet, these parameters remain constant, while their values vary across different positions along the sheet.<sup>27</sup>

The model illustrates a geometry (Plate) by modeling them on  $M^* = 0$  and  $\gamma = 0$ .

The most important physical characteristics, including skin friction factor and the Nusselt number, are provided in the following section after performing the proper similarity transformations.<sup>24</sup>

$$C_f^* = \frac{\ddot{A}_w}{\dot{A}_f u_w^2}, \quad Nu_r = \frac{l q_w}{k_f (T_w - T_\infty)}. \quad (13)$$

Here,  $\tau_w = \mu_{hnf} (1 - n_a) \frac{\partial u}{\partial y} \Big|_{y=0} + \mu_{hnf} \frac{n_a \Gamma}{\sqrt{2}} \left( \frac{\partial u}{\partial y} \right)^2 \Big|_{y=0},$

$$q_w = -k_{hnf} \frac{\partial T}{\partial y} \Big|_{y=0} + q_r \Big|_{y=0}. \quad (14)$$

Now, by substituting Eq. (14) into Eq. (13), the result is as follows:

$$C_f = \frac{x}{l} C_f^* = A_1 \left( 1 - n_a + \frac{n_a We}{2} f''(0) \right) f''(0), \quad N_r = - \left( A_6 + \frac{4}{3} Rd \right) \theta'(0). \quad (15)$$

### 3. ENTROPY GENERATION ANALYSIS

The expression for the entropy production in terms of dimensions is as follows:<sup>28,29</sup>

$$S_{gen} = \frac{k_f}{T_\infty^2} \left[ \frac{k_{hnf}}{k_f} + \frac{16 \sigma^* T_\infty^3}{3 k^* k_f} \right] \left( \frac{\partial T}{\partial y} \right)^2 + \frac{\mu_{hnf}}{T_\infty} \left( \frac{\partial u}{\partial y} \right)^2$$

$$\left( (1 - n_a) + \frac{n_a \Gamma}{\sqrt{2}} \frac{\partial u}{\partial y} \right) + \frac{\mu_{hnf}}{T_\infty K^*} u^2 + \frac{\sigma_f B_0^2}{T_\infty} u^2. \quad (16)$$

By applying Eq. (9) into Eq. (16), the entropy generation becomes

$$\begin{aligned}
 N_G &= \frac{S_{gen}}{(\Delta T k_f / T_\infty l^2)} \\
 &= \alpha_1 \left( A_6 + \frac{4}{3} Rd \right) (\theta')^2 \\
 &\quad + A_1 Br \left( f''^2 \left( (1-n_a) + \frac{Wen_a}{2} f'' \right) + K f'^2 \right) \\
 &\quad + M A_4 Br f'^2.
 \end{aligned} \quad (17)$$

### 3.1. Bejan number

In several scientific and engineering challenges, the temperature transfer entropy production to total entropy production ratio is essential. This is described as follows:<sup>29</sup>

$$\begin{aligned}
 Be &= \frac{\alpha_1 \left( A_6 + \frac{4}{3} Rd \right) (\theta')^2}{\alpha_1 \left( A_6 + \frac{4}{3} Rd \right) (\theta')^2 + A_1 Br} \\
 &\quad \left( f''^2 \left( (1-n_a) + \frac{Wen_a}{2} f'' \right) + K f'^2 \right) \\
 &\quad + M A_4 Br f'^2.
 \end{aligned} \quad (18)$$

## 4. GRAPHICAL RESULTS AND DISCUSSION

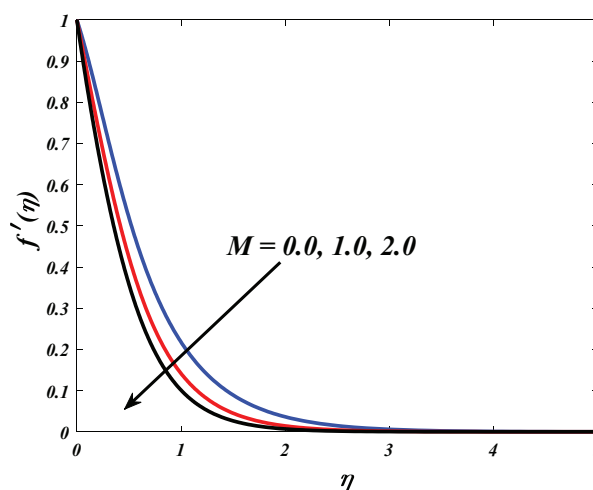
The collection of partial differential equations that represents Eqs. (1)–(3) and the corresponding conditions in Eq. (4) are transformed into a set of ODEs in Eqs. (10) and (11) with modified boundary conditions in Eq. (12) and are addressed by using the MATLAB program bvp4c solver. A key element of bvp4c is choosing a relatively low grade  $\eta_\infty$ . The step size and error tolerance in this investigation are  $h = 0.001$  and  $10^{-8}$ , respectively. The numerical findings indicate the effect of nondimensional factors using graphs and profiles. For various universal constant values,  $K = 0.1$ ,  $Rd = 0.1$ ,  $M = 2$ ,  $Ec = 0.1$ ,  $Q = 0.1$ ,  $Pr = 21$ ,  $Gr = 0.1$  and  $Br = 0.1$ . Table 1 shows the thermophysical properties of the base fluid and hybrid nanoparticles. The precise solutions are obtained from Table 2, which shows the variations in skin friction coefficient for different  $M$ . The results are found to be pretty accurate with previous results. Figures 2–18 depict the evaluation of hybrid nanofluid velocity, temperature, entropy production, Bejan number, skin friction and rate of heat transfer for different parameters.

**Table 1.** Thermophysical properties of base fluid blood, copper and silver hybrid nanofluid.<sup>30,31</sup>

| Physical properties                 | Blood                | Copper              | Silver              |
|-------------------------------------|----------------------|---------------------|---------------------|
| $\rho(\text{kgm}^{-3})$             | 1050                 | 8933                | 10500               |
| $C_p(\text{Jkg}^{-1}\text{K}^{-1})$ | 3617                 | 385                 | 235                 |
| $k_f(\text{Wm}^{-1}\text{K}^{-1})$  | 0.52                 | 400                 | 429                 |
| $\sigma(\Omega\text{m})^{-1}$       | 0.8                  | $5.96 \times 10^7$  | $6.3 \times 10^7$   |
| $\beta_T \times \frac{1}{K}$        | $1.8 \times 10^{-6}$ | $17 \times 10^{-6}$ | $18 \times 10^{-7}$ |
| Pr                                  | 21                   |                     |                     |

**Table 2.** Justification for  $-f''(0)$  different values of  $M$  when  $K = Gr = M^* = 0$ .

|   | Chu et al. <sup>32</sup> | Akbar et al. <sup>33</sup> | Ali et al. <sup>34</sup> | Rekha et al. <sup>23</sup> | Present results MATLAB (bvp4c) |
|---|--------------------------|----------------------------|--------------------------|----------------------------|--------------------------------|
| 0 | 1.0000                   | 1.0000                     | 1.0000                   | 1.000483                   | 1.000246                       |
| 1 | 1.41423                  | 1.41421                    | 1.4142                   | 1.414216                   | 1.414219                       |
| 5 | 2.44946                  | 2.44948                    | 2.4495                   | 2.449489                   | 2.449620                       |



**Fig. 2.** (Color Online) Influence of  $M$  on  $f'(\eta)$ .

### 4.1. Discussion on velocity profile

Figures 2–5 are included herein to visually depict the alterations observed in the velocity profile of a hybrid nanofluid across different magnetic field parameter ( $M$ ), porous permeability ( $K$ ), Grashof number ( $Gr$ ) and Weissenberg number ( $We$ ) for both  $M^* = 0$  and  $\gamma = 0$  conditions. Figure 2 depicts the effect of the magnetic parameter ( $M$ ), on the velocity profile for Blood/Ag+Cu hybrid nanofluid mixture combinations. The relationship between the  $M$

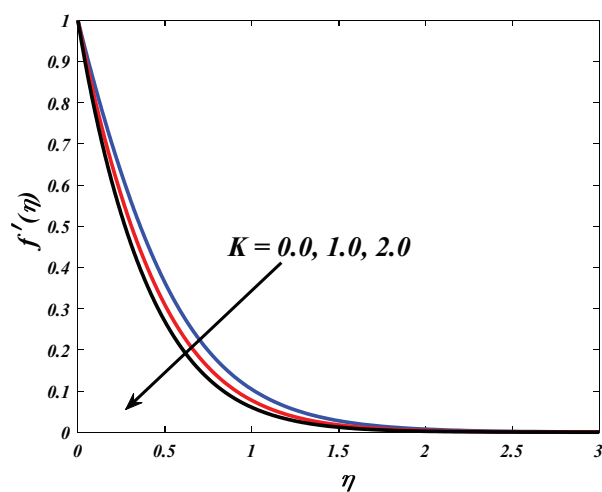


Fig. 3. (Color Online) Influence of  $K$  on  $f'(\eta)$ .

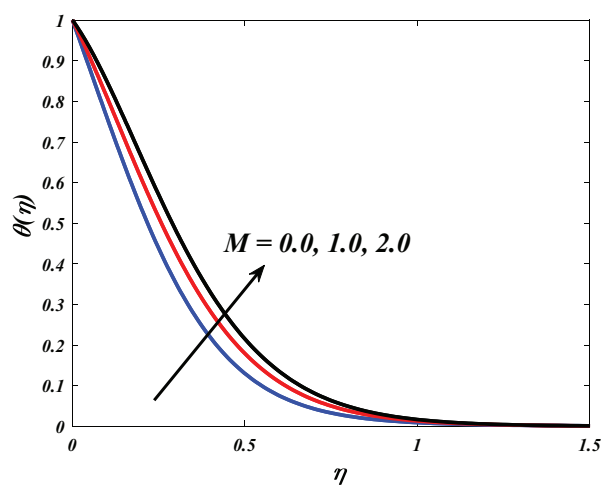


Fig. 6. (Color Online) Influence of  $M$  on  $\theta(\eta)$ .

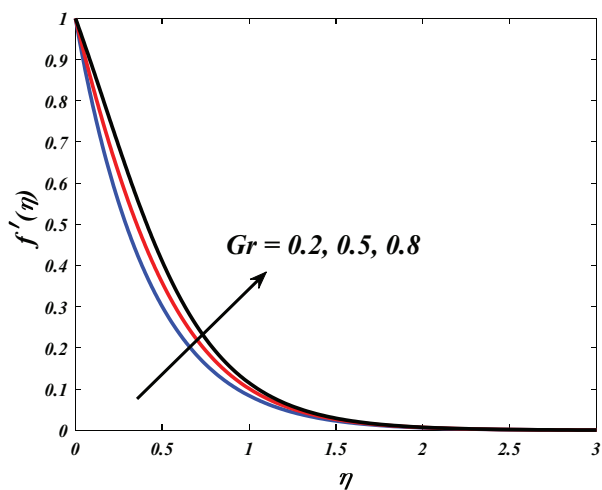


Fig. 4. (Color Online) Influence of  $Gr$  on  $f'(\eta)$ .

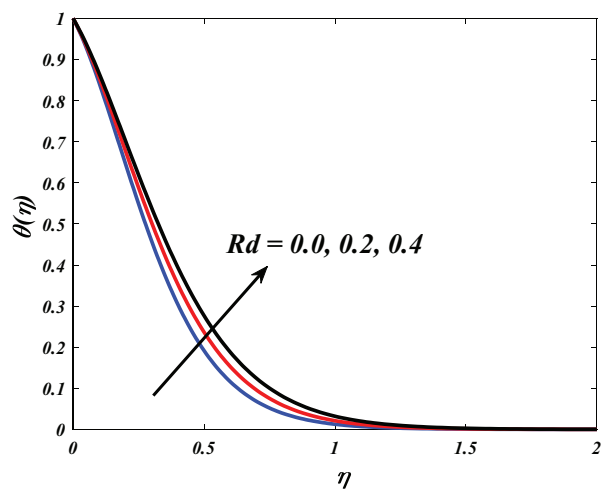


Fig. 7. (Color Online) Influence of  $Rd$  on  $\theta(\eta)$ .

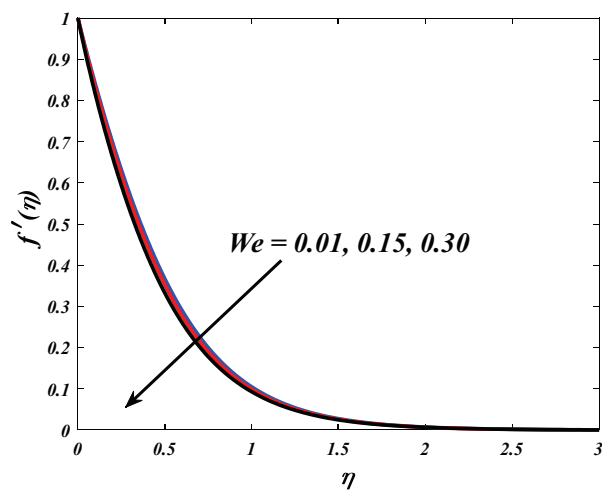


Fig. 5. (Color Online) Influence of  $We$  on  $f'(\eta)$ .

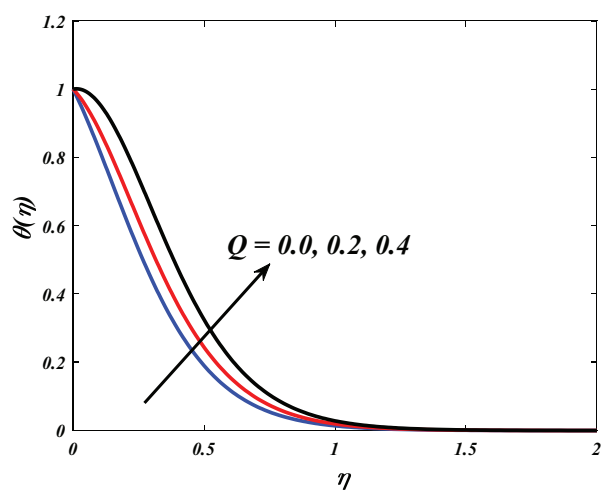
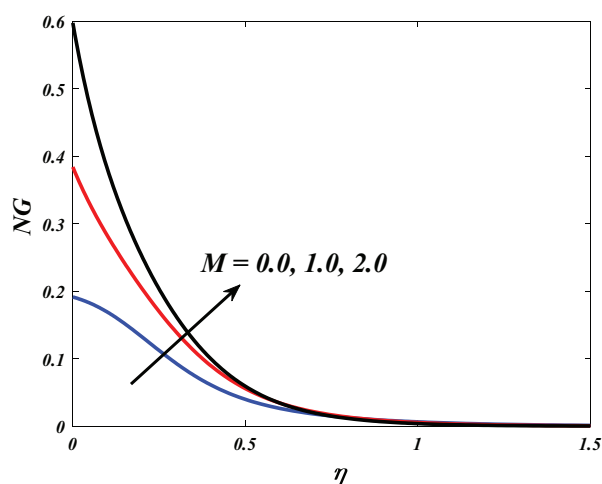
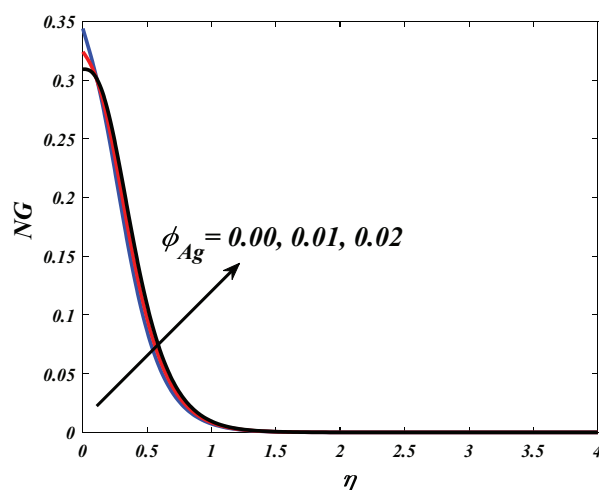
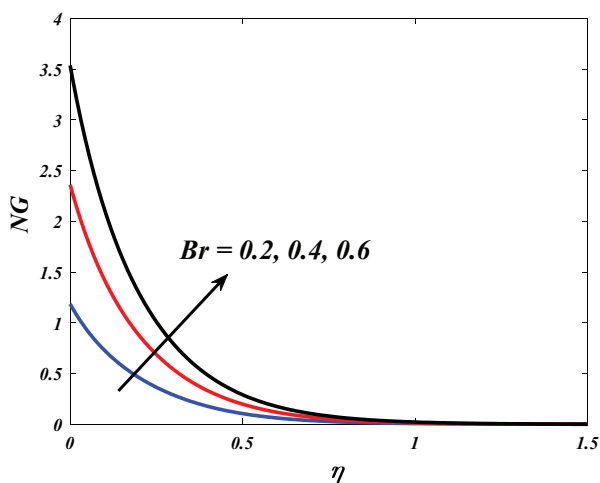
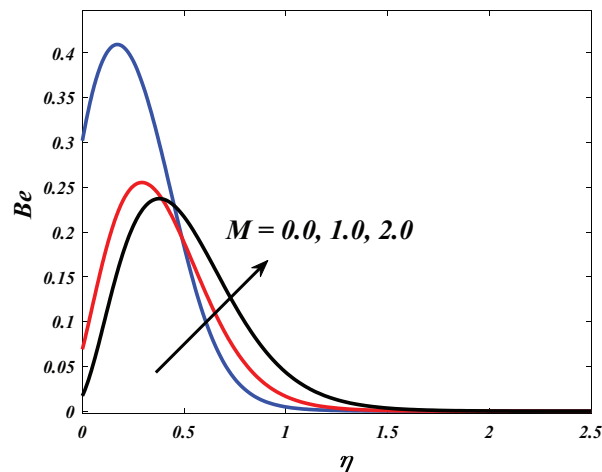
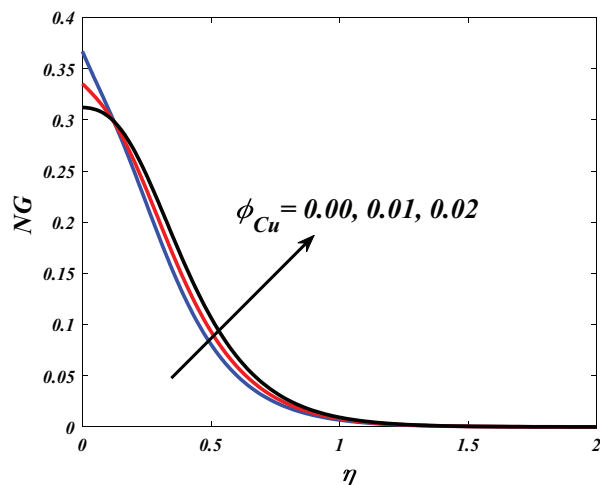
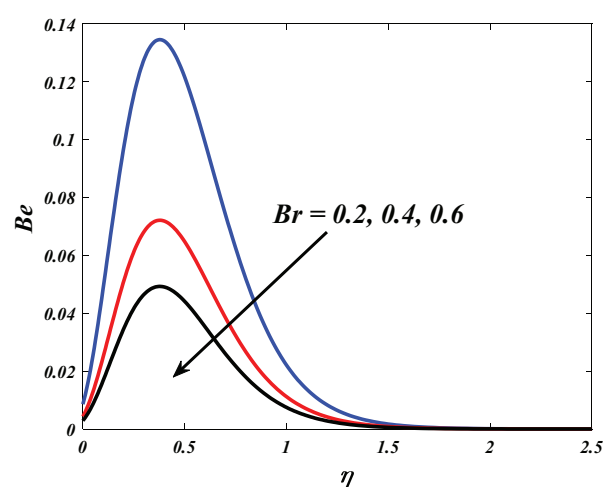


Fig. 8. (Color Online) Influence of  $Q$  on  $\theta(\eta)$ .



Fig. 9. (Color Online) Influence of  $M$  on  $NG$ .Fig. 12. (Color Online) Influence of  $\phi_{Ag}$  on  $NG$ .Fig. 10. (Color Online) Influence of  $Br$  on  $NG$ .Fig. 13. (Color Online) Influence of  $M$  on  $Be$ .Fig. 11. (Color Online) Influence of  $\phi_{Cu}$  on  $NG$ .Fig. 14. (Color Online) Influence of  $Br$  on  $Be$ .

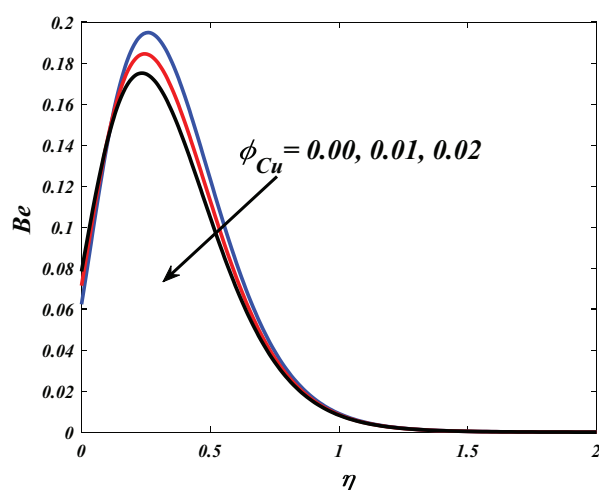


Fig. 15. (Color Online) Influence of  $\phi_{Cu}$  on Be.

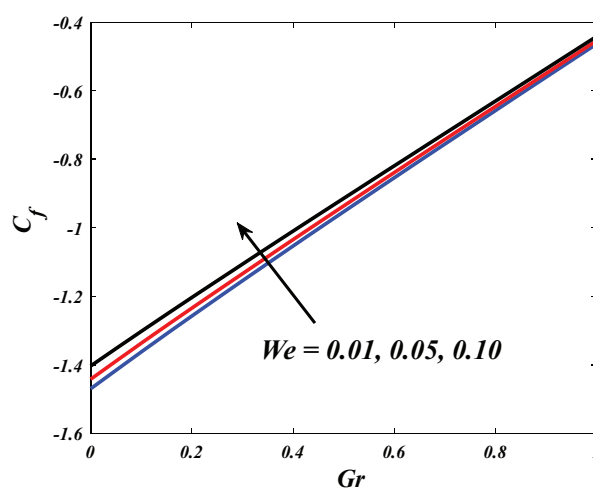


Fig. 18. (Color Online) Influence of We and Gr on  $C_f$ .

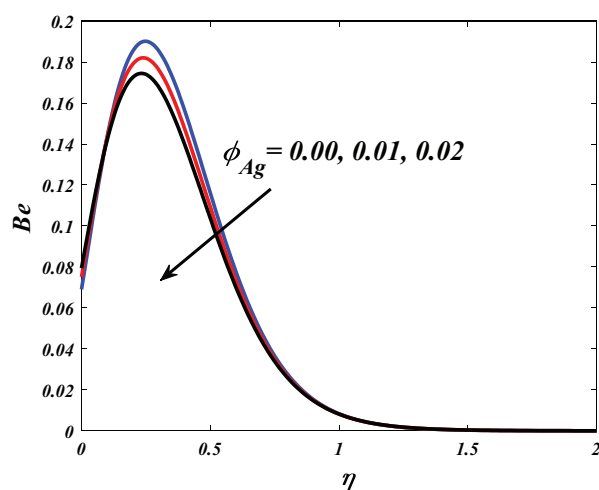


Fig. 16. (Color Online) Influence of  $\phi_{Ag}$  on Be.

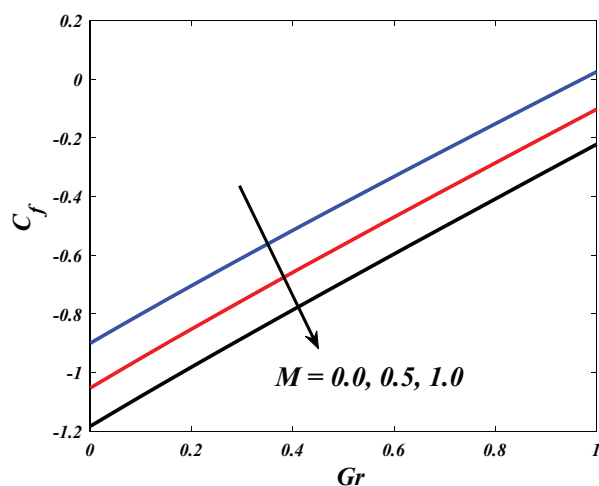


Fig. 17. (Color Online) Influence of  $M$  and Gr on  $C_f$ .

increase and the fluid velocity decrease has been established. The findings are consistent with the concept that the resistive force is a crucial factor in guiding and decelerating the movement of fluids. The Lorentz force, a resistive field and a magnetic parameter are intricately interconnected in the physical realm. The impact of porosity ( $K$ ) on dimensionless velocity is depicted in Fig. 3. With an increase in the permeability parameter, there is a slight growth in the thickness of the momentum boundary layer. With increased porosity, i.e. bigger  $K$ , the fluid has more room for flow and, consequently, its velocity increases. Figure 4 illustrates the alteration in the velocity profile resulting from an augmentation in the Grashof number within said profile. While temperature variations are the main source of buoyant forces, the Grashof number may have an impact on the flow's velocity profile. The buoyant forces are said to outweigh the viscous forces when the Grashof number is larger. Large-scale flow structures and greater fluid motion result from the buoyancy forces' supremacy. Figure 5 illustrates the impact of elevated Weissenberg numbers ( $We$ ) on the velocity profile. An increase in the Weissenberg number ( $We$ ) leads to a reduction in the velocity profile. Both the viscous and elastic characteristics of the fluid have an impact on its behavior. The Weissenberg number serves to quantify the ratio of elastic effects to viscous effects present within a given flow. A greater dominance of the elastic properties is indicated by a higher Weissenberg number. Consequently, it would be inaccurate to assert that elevated Weissenberg numbers invariably result in a reduction of the velocity profile.

## 4.2. Discussion on temperature profile

Figures 6–8 show the effects of the parameters  $M$  for the magnetic field,  $R_d$  for radiation and  $Q$  heat production on the temperature of the hybrid nanofluid. Figure 6 demonstrates the observation of temperature increase in the hybrid nanofluid for different magnetic parameter values of the field denoted as  $M$ . The presented figure demonstrates that an increase in the magnetic parameter of the field results in an elevation of the temperature profile. The velocity profile presented above indicates that the mobility of the fluid is diminished due to an escalation in the magnetic field parameter. The gradual movement of the fluid results in the conversion of its kinetic energy into thermal energy, leading to an increase in temperature. Furthermore, an examination is being conducted to determine the effect of an elevated magnetic field on the reduction of the thermal and momentum boundary layer of the hybrid nanofluid. An inverse relationship has been observed between the magnetic parameter and the density of the hybrid nanofluid. The augmentation of the magnetic field parameter induces a reduction in fluid density, leading to a rise in temperature. Figure 7 demonstrates the impact of the radiation parameter  $R_d$  on the fluid temperature profile. It is observed that higher values of  $R_d$  lead to an enhancement in the temperature profile. The augmentation in temperature of the fluid is the underlying factor that leads to an escalation in the thickness of the boundary layer. The rise in temperature of the hybrid nanofluid can be attributed to the enhanced effectiveness of the radiation parameter in heating the fluid. Moreover, the temperature of the hybrid nanofluid increases due to the enhancement of thermal diffusivity of the fluid, which is attributed to higher values of the radiation parameter. Figure 8 analyzes the influence of the temperature of the fluid graph across different magnitudes of the heat generation parameter  $Q$ . The observation indicates a positive correlation between the increase in heat generation  $Q$  and the corresponding rise in the temperature profile. Elevated levels of  $Q$  lead to an increase in the internal energy of liquid particles, thereby resulting in a rise in temperature.

## 4.3. Discussion on entropy generation

The local entropy generation dispersion for a range of  $M$ ,  $Br$ ,  $\phi_{Cu}$  and  $\phi_{Ag}$  values is shown in Figs. 9–12. The influence of varying proportions of the magnetic field on the NG is seen in Fig. 9. It is acknowledged that the NG increases when the magnetic field parameter  $M$  increases. The phenomenon of energy losses can be

attributed to the interaction between the fluid and the magnetic field. The aforementioned losses may arise from Joule heating, a phenomenon whereby electrical energy is transformed into heat due to the electrical currents that are generated by the magnetic field. The phenomenon of Joule heating is known to augment the generation of entropy and can be especially noteworthy in fluids that exhibit high conductivity. The impact of the Brinkman number parameter ( $Br$ ) on changes in entropy generation NG is shown in Fig. 10 correspondingly. This graph shows that  $Br$  increases as the NG intensifies. The impact of Brinkman number on the velocity and temperature fields is contingent upon the particular flow conditions and medium properties. The modifications in fluid heat and friction transfer ensure that they have the potential to impact the rate of entropy generation. The data that were gathered and displayed in Figs. 11 and 12 suggest that a rise in the volume percentage of nanoparticles  $\phi_{Cu}$  and  $\phi_{Ag}$  causes an acceleration in the pace at which entropy is formed. This is shown by the correlation between the two sets of figures. Nanoparticles may prevent heat transport inside the fluid because of their high surface area-to-volume ratio. An increase in nanoparticle volume fraction results in a corresponding increase in the number of nanoparticles present in the fluid, thereby causing an increase in the number of obstacles that impede heat transfer. The heightened resistance to heat transfer produces more irreversibilities, thereby contributing to an increase in entropy generation.

## 4.4. Discussion on Bejan number

The influence of the Bejan number is seen in Figs. 13–16 for various levels of  $M$ ,  $Br$ ,  $\phi_{Cu}$  and  $\phi_{Ag}$ . Investigated from Fig. 13 is the Bejan number falloff for increasing magnetic parameter values. Entropy creation increases physically, which causes Bejan number to drop. As seen in Fig. 14, the Bejan number decreases as  $Br$  increases. This is because the other factors, such as fluid friction, are immutable in each instance, overshadowing the irreversibility of thermal and mass transfer. Figures 15 and 16 show how the Bejan number decreases with increasing nanoparticle volume fractions  $\phi_{Cu}$  and  $\phi_{Ag}$ .

## 4.5. Discussion on skin friction and heat transfer

Figure 17 endorses the impression of magnetic filled  $M$  and Grashof number  $Gr$  parameters on  $C_f$ . It is

recognized that both magnetic field  $M$  and Gr parameters decrease to the skin friction. The physical rationale behind the decrease in skin friction with higher values of both the magnetic field and Gr can be attributed to their distinct effects on fluid flow behavior. In the case of an elevated magnetic field, as encountered in MHD flows, the Lorentz force induced by the magnetic field interacts with the moving fluid, suppressing turbulence and dampening velocity fluctuations near the surface. This reduction in velocity gradients along the boundary layer leads to a decrease in skin friction. On the other hand, with a higher Gr, which signifies a stronger buoyancy-driven flow due to temperature differences, convective motion becomes more pronounced, enhancing fluid mixing and increasing velocity gradients near the surface. This heightened convective activity diminishes the energy dissipation within the boundary layer, resulting in lower skin friction. Figure 18 is outlined to reveal the influence of We and Gr on  $C_f$ . It is revealed that slightly decreases on the  $C_f$  of hybrid nanofluid over the plate on the We parameter and the same nature is observed in the  $C_f$  parameter. Higher values of We and Gr lead to decreased skin friction due to distinct underlying mechanisms. In the case of the We, which characterizes viscoelastic behavior, elevated values indicate a stronger dominance of elastic forces over viscous forces in materials undergoing deformation. This increased elasticity reduces the tendency of the fluid near the surface to adhere and resist motion, resulting in reduced friction along the boundary. Similarly, higher Grs imply enhanced buoyancy-driven fluid motion caused by temperature differences, which promotes mixing and reduces the thickness of the thermal boundary layer near the surface. Figure 19 endorses the impression of magnetic field ( $M$ ) and Gr parameters on  $N_r$ . It is recognized that both parameters ( $M$ ) and (Gr) decrease to the rate of heat transfer. Due to different physical reasons, higher magnetic fields and Grs reduce heat transmission. In MHD flows with a greater magnetic field, the Lorentz force limits fluid velocity and convective heat transfer. Reduced fluid velocity near the surface slows convection heat transfer. However, high Gr values suggest temperature-driven buoyancy-driven convection. Convective motion improves heat transmission but thins the thermal boundary layer at the surface. This smaller boundary layer lowers heat conduction across the material by lowering the fluid–solid temperature differential. Thus, larger magnetic fields and Gr numbers reduce fluid motion and thermal boundary layers, reducing convection and conduction heat transfer rates. Figure 20 endorses the impression of heat source ( $Q$ ) and (Gr)

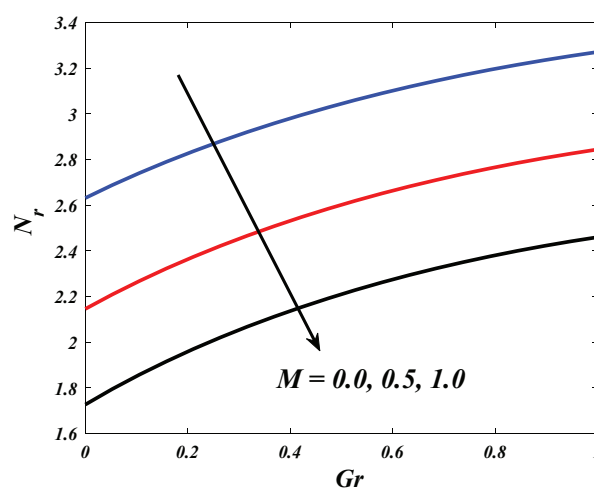


Fig. 19. (Color Online) Influence of  $M$  and Gr on  $N_r$ .

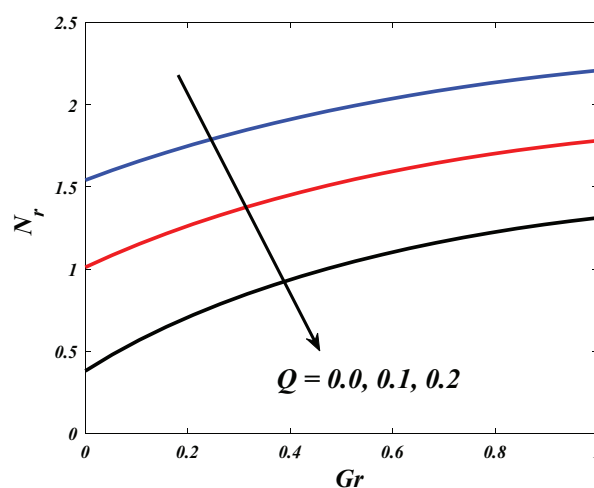


Fig. 20. (Color Online) Influence of  $Q$  and Gr on  $N_r$ .

parameters on  $N_r$ . It is recognized that both ( $Q$ ) and (Gr) parameters decrease to the rate of heat transfer. Heat transfer rates decrease with higher heat source and Grs owing to various physical reasons. The temperature gradient increases near an elevated heat source, such as fluid heat production. This temperature differential increases buoyancy-driven convection currents, increasing heat transfer and lowering the surface's thermal boundary layer. This smaller barrier layer reduces heat conduction and heat transfer by lowering the fluid–solid temperature differential. Due to temperature differences, higher Grs increase buoyancy effects and convective heat transfer. However, the thermal boundary layer thins, lessening the temperature difference across the layer and decreasing the transfer of heat. Figure 21 endorses the impression of radiation (Rd) and

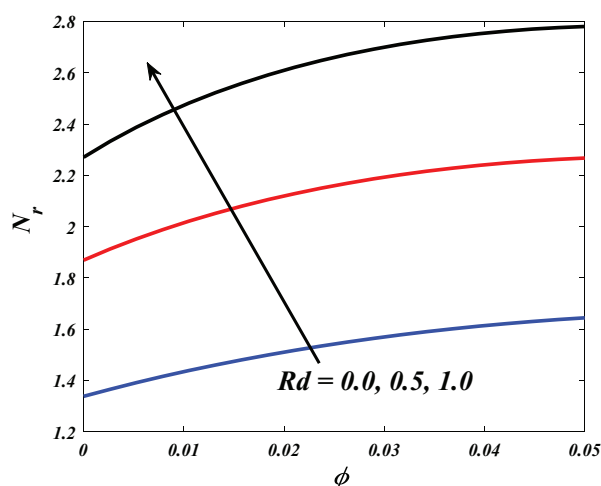


Fig. 21. (Color Online) Influence of  $Rd$  and  $\phi$  ( $=\phi_{Cu} + \phi_{Ag}$ ) on  $N_r$ .

nanoparticle volume fraction parameters ( $\phi = \phi_{Ag} + \phi_{Cu}$ ) on  $N_r$ . It is recognized that  $N_r$  of hybrid nanofluid at the surface increases for amplifying radiation ( $Rd$ ) and nanoparticle volume fraction ( $\phi$ ) parameters. Physically, higher radiation intensity increases radiative heat transfer through electromagnetic waves. Radiation helps heat exchange even when conduction and convection are restricted. Nanoparticles' efficient energy transport capabilities promote heat transfer in nanofluids with higher nanoparticle volume fractions. Nanoparticles transfer and convect heat better than the base fluid, improving thermal conductivity and convection. This efficiency boosts nanofluid heat transfer rates.

## 5. CONCLUSIONS

The objective of this investigation is to analyze entropy generation in the context of radiative heat transfer under the influence of a magnetic field over a permeable plate, specifically focusing on the flow of a hybrid nanofluid consisting of Ag+Cu-Blood. A volume fraction nanofluid approach was used to investigate the behavior of Ag+Cu nanoparticles dispersed into the blood. It's important to note that the non-Newtonian nanofluid model takes into account the presence of blood. Various factors play a significant role in this study, including the particle fluid parameter, heat source, magnetic field parameter, porous medium, and radiation parameter. The nonlinear equations for velocity and energy were solved by utilizing the MATLAB program bvp4c solver. The findings are provided in the form of tables and graphs that only have two dimensions. The following key outcomes were obtained from this investigation:

- The velocity profile of MHD tangent hyperbolic hybrid nanofluids near a porous plate increases with greater values of the Grashof number, indicating an upward trend in the velocity.
- The velocity of MHD tangent hyperbolic hybrid nanofluids decreases as the  $We$  increases, showing an inverse relationship between velocity and  $We$ .
- The temperature of MHD tangent hyperbolic hybrid nanofluids rises with higher values of the heat source and radiation parameters, indicating the impact of these parameters on the thermal behavior of the system.
- Increasing the volume percentage of nanoparticles in MHD tangent hyperbolic hybrid nanofluids accelerates the rate of entropy generation, suggesting that nanoparticle concentration plays a significant role in entropy production.
- Both higher magnetic field strength and Grashof number values contribute to the reduction of skin friction in MHD tangent hyperbolic hybrid nanofluids near a porous plate, and there is a negative correlation between these factors and skin friction.
- The rate of heat transfer in MHD tangent hyperbolic hybrid nanofluids is enhanced by increasing nanoparticle volume fraction and radiation parameters, indicating the potential of these factors to improve heat transfer efficiency in the system.

The research findings significantly contribute to enhancing our comprehension of the behavior of the hybrid nanofluid subjected to radiative heat transfer and a magnetic field above a permeable plate. Nevertheless, like any scientific study, there exist certain limitations and potential avenues for future exploration. Specifically, this study is limited by the absence of experimental validation and the neglect of complex nanoparticle interactions. Moving forward, an exciting prospect lies in the incorporation of multi-scale modeling techniques to explore the dispersion of nanoparticles within the nanofluid. By considering phenomena such as Brownian motion, thermophoresis and nanoparticle aggregation, a more comprehensive understanding of the nanofluid's behavior, particularly in intricate physiological settings, can be attained.

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## CONFLICT OF INTEREST

The authors declared no potential conflicts of interest with respect to the research, authorship and/or publication of this paper.

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